High school · optimisation and olympiads

Inequalities and equality cases

Lesson and method

Before you begin

Squares, factorisation and square roots.

Ideas and method

For every real $u, u^2 \geq 0$, with equality exactly when $u = 0$. Many inequalities reduce to this observation.

For $a, b \geq 0, (\sqrt{a} - \sqrt{b})^2 \geq 0$ gives $a + b \geq 2\sqrt{ab}$, the arithmetic–geometric mean inequality. Equality holds exactly when $a = b$.

Proving a bound is not enough to announce a maximum: an admissible equality case must attain it. Positivity and domain conditions are part of the argument.

Worked example

If $a + b = 10$ with $a, b \geq 0$, then $(a - b)^2 \geq 0$ gives $100 - 4ab \geq 0$, so $ab \leq 25$. The pair $a = b = 5$ attains 25, proving that it is the maximum.

Explore

Plot $f(x) = x(10 - x)$ for $0 \leq x \leq 10$ and move the point. Then prove the maximum without relying on the picture.